

Two-Stage Water Rocket & Its Simulation Platform

Transcultural & transectoral collaboration in transdisciplinary education

A bilingual STEM unit bridging physics, engineering and AI — co-delivered across cultures and sectors

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Significance — Why it matters

How do you teach abstract mechanics to young children across a language and sector boundary? We designed a **bilingual (Chinese–English)** STEM unit and delivered it as a **cross-regional service-learning project** — Hong Kong university students teaching primary-school children in mainland China. The water rocket becomes a safe, low-cost “living laboratory” for force, pressure and acceleration.



Transdisciplinary
physics (mechanics) × engineering design × computer science (AI) × pedagogy

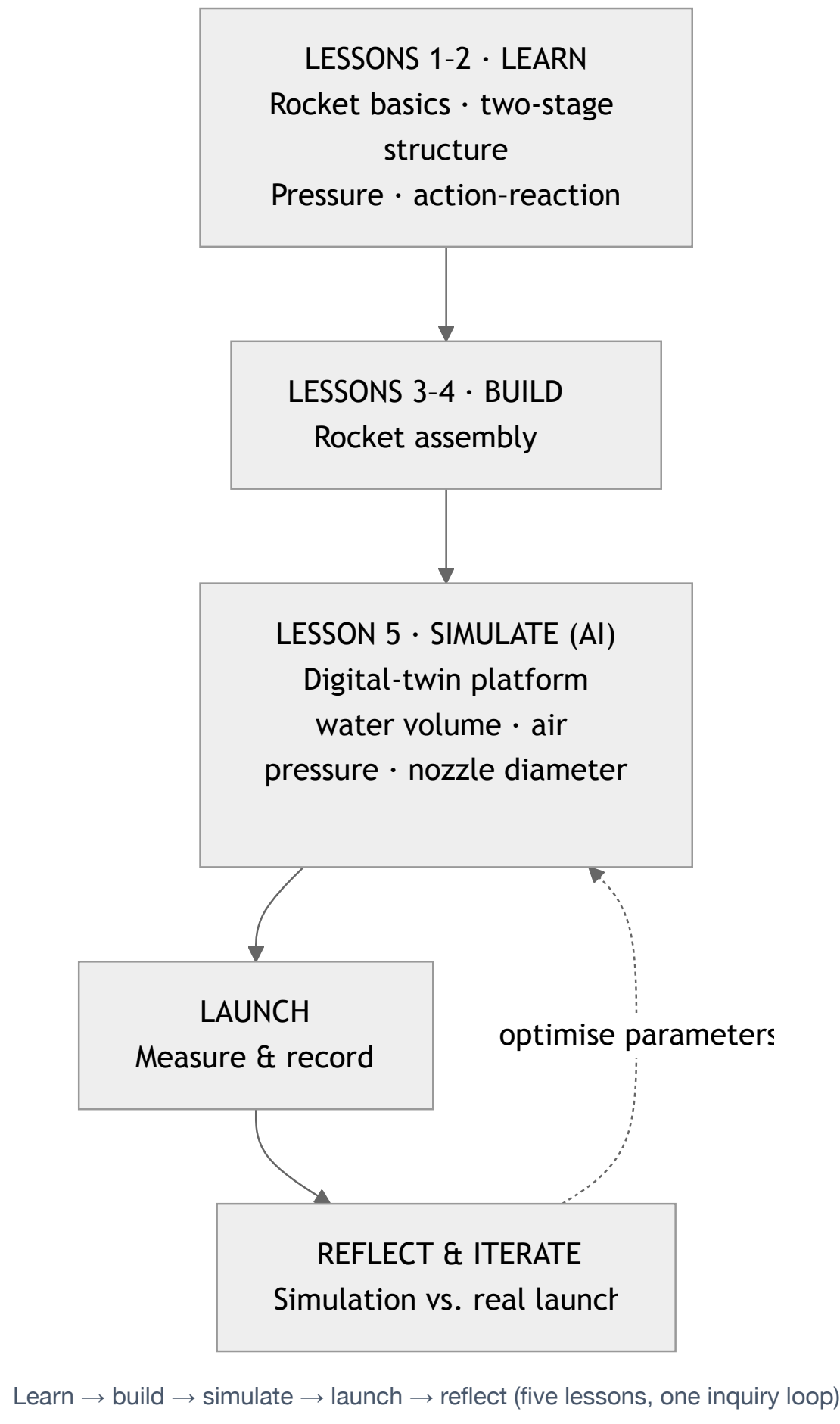
Transectoral
university ↔ primary school ↔ community — education, technology and social service

Transcultural
Hong Kong ↔ Jiangxi (Anyuan / Sanbaishan); bilingual delivery and cross-cultural exchange

Objectives

Knowledge Students can explain launch principles, pressure vs. air pressure, action–reaction, acceleration and air resistance.	Ability Students practise hypothesis → test → optimise ; variable control; data recording; engineering design.	Transdisciplinary Model a pedagogy that couples AI simulation with hands-on experiment , transferable to other STEM topics.
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Approach & Methodology



A custom **AI simulation platform** acts as a **pre-launch “digital twin”**: students sweep **water volume**, **air pressure** and **nozzle diameter** to predict the optimal launch *before* the physical test — reducing cost and safety risk while grounding intuition in theory.

Delivered **bilingually (Chinese–English)** across five scaffolded lessons, co-taught by university students for primary-school learners.



Key Findings & Implications

Anticipated outcomes	By the end of the unit, a target of [X] % of learners are expected to explain action–reaction and pressure; simulation predictions will be validated against [M] real launches.
Stakeholder impact	Learners grasp core mechanics and build inquiry skills; the school & community gain a reusable, low-cost STEM module; we, as facilitators , develop transdisciplinary teamwork and cross-cultural communication.
Next steps	Map every physics concept one-to-one onto build/launch steps; add academic-standard scenarios; build a student innovation zone to close the loop learn → practise → create .



“The children connected pressure and force faster than any worksheet ever did.”
— **Teacher testimony (to be confirmed)**

“I learned why my rocket flew higher when I pumped in more air.”
— **Student testimony (to be confirmed)**

References & Acknowledgement

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